Skillsheet E | ECON 0100 | Fall 2026

Sellers and market power.

Use this Skillsheet to guide your studying. You'll find a list of every skill in Part E, what practice unlocks it, and the standard you must meet to pass it on a Checkpoint.

Use the three types of practice to identify to yourself whether you understand the skill. Do each practice without much support from the solutions or other resources. This will tell you whether you understand the skill, and if not, it will give you a venue to get yourself unstuck. Do not simply submit the correct answers: since practice is graded for completion not correctness, there is no reason to submit perfect answers you do not fully understand.

Each skill has three types of practice: the **Exercise** (done together in class), the **Vignette** (done together in recitation), and the **Homework** (done on your own time, due Sundays).

- Complete **2 of 3** practices for a skill to unlock the skill on Checkpoint E.
- Complete **3 of 3** practices for a skill to unlock the Reattempt.

Code	Skill	Practice		
E1.1	Production Functions	Exercise E1	Vignette E1	HW E1
E1.2	Cost Curves	Exercise E1	Vignette E1	HW E1
E2.1	The Competitive Firm	Exercise E2	Vignette E2	HW E2
E2.2	Entry, Exit & the Long Run	Exercise E2	Vignette E2	HW E2
E3.1	Monopoly Price & Quantity	Exercise E3	Vignette E3	HW E3
E3.2	Market Power & Deadweight Loss	Exercise E3	Vignette E3	HW E3
E4.1	Best Responses in a Duopoly	Exercise E4	Vignette E4	HW E4
E4.2	Duopoly Equilibrium	Exercise E4	Vignette E4	HW E4

E1.1 | Production Functions

A production function tells us how much output a firm gets from its inputs, and so how many inputs each unit of output requires.

Standard. You pass this skill if you can:

- Read a production function, as a table or a graph, in both directions: the output from a given amount of labor, and the labor a given output requires.
- Calculate the marginal product of each added worker and identify diminishing marginal product.
- Turn labor into cost with a wage and a fixed cost: calculate variable cost and total cost at each output.

E1.2 | Cost Curves

Average and marginal costs are built from total cost, and each one answers a different question about the firm.

Standard. You pass this skill if you can:

- Move among the cost measures in a table: fixed, variable and total cost, the three averages (AFC, AVC, ATC), and marginal cost.
- Use $ATC = AVC + AFC$ and $MC = \Delta TC/\Delta Q$ to fill in a missing value.
- Decide whether ATC is falling or rising at a quantity by comparing it with MC, and locate the lowest point of ATC where the two are equal.
- Match a firm's cost structure (high or no fixed cost, constant or rising marginal cost) to its ATC and MC curves.

E2.1 | The Competitive Firm

A firm with no market power takes the price as given and chooses only how much to make.

Standard. You pass this skill if you can:

- Find a price taker's marginal revenue: it equals the market price.
- Find the profit-maximizing quantity where price equals marginal cost.
- Calculate profit or loss as $(P - ATC) \times q$.
- Compare the loss from operating with the loss from shutting down this period, and choose between them.

E2.2 | Entry, Exit & the Long Run

Profit draws firms in and losses push them out, until firms earn zero economic profit.

Standard. You pass this skill if you can:

- Decide from a firm's profit or loss whether firms enter or exit, and which way market supply and the price move.
- Find the long-run price at the lowest point of ATC, where economic profit is zero.
- After a change that hits every firm, such as a per-unit tax, a subsidy, or a shift in demand, give the long-run price, each firm's output, market quantity, and whether the number of firms rises or falls.

E3.1 | Monopoly Price & Quantity

A monopolist faces the whole market's demand, so selling one more unit lowers the price on every unit.

Standard. You pass this skill if you can:

- Find the monopolist's quantity where marginal revenue equals marginal cost.
- Find the price on the demand curve at that quantity, not on marginal revenue.
- Calculate the monopolist's profit, and show it on a graph.

E3.2 | Market Power & Deadweight Loss

A seller with market power sells less than the efficient quantity.

Standard. You pass this skill if you can:

- Find the efficient quantity where demand (marginal benefit) meets marginal cost.
- Calculate the deadweight loss of a monopoly as the triangle between demand and marginal cost, from the monopoly quantity to the efficient quantity.
- Compare price and quantity under monopoly with the competitive outcome in the same market.

E4.1 | Best Responses in a Duopoly

With two sellers, each one's best quantity depends on how much the other sells.

Standard. You pass this skill if you can:

- Find a seller's best response to a given quantity from its rival by setting its marginal revenue equal to marginal cost.
- Write a seller's best response as an equation in its rival's quantity.
- Find each seller's best responses and the Nash equilibrium when the two choose between a high and a low output in a payoff matrix.

E4.2 | Duopoly Equilibrium

In a Nash equilibrium both sellers are choosing a best response to each other.

Standard. You pass this skill if you can:

- Solve two best responses together for each seller's Nash equilibrium quantity, using symmetry when the sellers are identical.

- Find the market quantity, the price from demand, and total profit.
- Compare the duopoly's price and quantity with the monopoly and competitive outcomes in the same market.